Macroscale conjugated microporous polymers: controlling versatile functionalities over several dimensions

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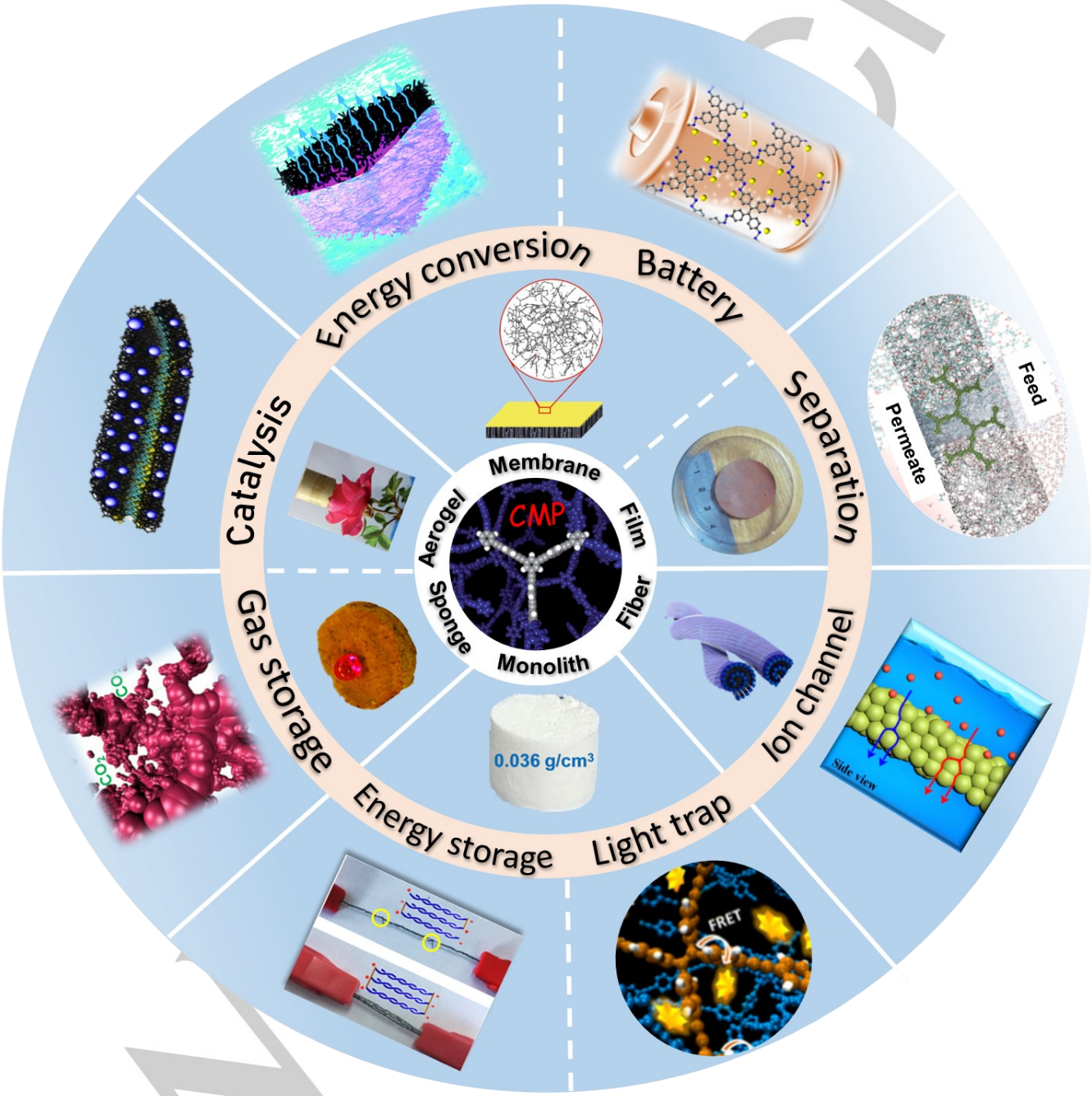

Abstract: Since discovered in 2007, conjugated microporous polymers (CMPs) have been developed for numerous applications including gas adsorption, sensing, organic and photo-redox catalysis, energy storage, *etc.* While featuring abundant micropores, the structural rigidity derived from CMPs' stable π-conjugated skeleton leads to insolubility and thus poor processability, which severely limits their applicability, *e.g.* in CMP-based devices. Hence, the development of CMPs whose structure can not only be controlled on the micro- but also on the macroscale have attracted tremendous interest. In conventional synthesis procedures, CMPs are obtained as powders, but in recent years various bottom-up synthesis strategies have been developed, which yield CMPs as thin films on substrates or as hybrid materials, allowing to span length scales from individual conjugated monomers to micro-/macro-structures. This review surveys recent advances on the construction of CMPs into macroscale structures, including membranes, films, aerogels, sponges, and other architectures. Our focus is to describe the underlying fabrication techniques and the implications which follow from the macroscale morphologies, involving new chemistry and physics in such materials for applications like molecular separation/filtration/adsorption, energy storage and conversion, photothermal transformation, sensing or catalysis.

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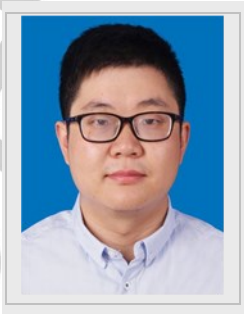
1. Introduction



Weiyi Zhang obtained his Ph.D. degree at 2017 from the Max Planck Institute of Colloids and Interfaces under the supervision of Prof. J. Yuan and Prof. M. Antonietti. Then he pursued his post-doc research at Clarkson University, U.S. (with Prof. J. Yuan), and Stockholm University, Sweden (with Prof. J. Yuan). In 2020 he started independent research career as a full professor at the College of Materials Science and Engineering, Donghua University, China. His research focuses on functional porous polymers and polyelectrolytes in particular for environmental and energy applications.

Yaozu Liao received his Ph. D degree in 2011 from Tongji University (with Prof. Xin-Gui Li), during which he also worked as a visiting scholar at University of California, Los Angeles (with Prof. Richard Kaner). Then after worked as a lecturer at University of Shanghai for Science and Technology, he continued his Marie Curie and Alexander von Humboldt post-doc research at University of Bristol, UK, and Technical University of Berlin, Germany. In 2015, Yaozu Liao joined State Key Laboratory for Modification of Chemical Fibers and Polymer Materials and Donghua University as a full professor. His research focuses on functional fibers and porous polymers.

Porous organic polymers (POPs)[1] are an emerging class of multifunctional porous materials, which are constructed from covalently bonded organic moieties. Because of their unique properties derived from their tunable porous structures and the multiple functional groups, which can be included in their backbones, POPs have been actively investigated for multiple applications including gas storage/separation[2, 3], catalysis[4-6], optoelectronics[7, 8], sensing[9-11], energy storage and conversion [12, 13]. Aside from their high specific surface areas, they furthermore show excellent chemical stability, light-weight and a versatile chemistry for modification and functionalization. POPs can be further classified into two categories based on their crystallinity. Crystalline POPs are usually summarized under the name covalent organic frameworks[14-16] (COFs). Amorphous POPs are further divided, based on their structure or construction principles into hyper-crosslinked polymers[17, 18] (HCPs), polymers of intrinsic microporosity[19, 20] (PIMs), porous aromatic frameworks[21, 22] (PAFs) and others. Recent research has seen increasing interest on amorphous POPs, especially when their skeleton is π-conjugated. In these highly porous networks, π-conjugation is superimposed with meso-/micro-porosities, *i.e.* electron transport in the polymer backbone is accompanied by mass transport in the porous system, which enable many intriguing properties and applications, *e.g.* in energy storage and conversion[23, 24], or optoelectronics[25, 26]. As such, the importance

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of π-conjugation in porous polymer networks has defined another emerging functional POP class — the conjugated microporous polymers (CMPs). As mentioned, such CMPs[27] are a unique class of POPs exhibiting extended π-conjugated structures and permanent nanopores. Earlier, McKeown *et al.*[28] discussed an important concept of preparing robust nanoporous materials by the covalent binding of planar molecules *via* a rigid spirocyclic linker. In 2007, Cooper *et al.* [29] reported a highly cross-linked, microporous poly(arylene ethynylene)s network, thus the first microporous polymer network with distinct π-conjugation and introduced the term "conjugated microporous polymer (CMP)" for this class of materials. Since their discovery, CMPs chemistry has intrigued scientists across the globe and been promoted rapidly, resulting in a strong growth in publications over the last decade. For instance, Thomas *et al.* [30] reported a spirobifluorene-type CMP with stable interface and application potential in organic light emitting diodes and organic solar cells. Jiang *et al.*[31] used a carbazole derivative, triindolocarbazole (TCB), to synthesize a TCB-CMP that exhibits blue luminescence and could be used to detect arenes upon exposure to their vapors. Thereafter, the CMP family has been largely developed and utilized in a plethora of fields including light emitters[32], energy storage[33, 34], catalysis[35], gas storage/separations[36], chemosensors[37, 38], energy conversion[39, 40], biological applications[41], *etc.*

CMPs are usually formed *via* C-C or C-N coupling reactions, including the Sonogashira–Hagihara [42, 43], Suzuki-Miyaura [44, 45] or Yamamoto coupling reactions[46-49], oxidative polymerizations[50, 51], Schiff base reactions[52-55], cyclotrimerizations[56, 57], phenazine ring fusion reactions[58], Buchwald-Hartwig Amination[59-64] and others (**Table 1**). Notably, Liao *et al.* [65-68] introduced Chichibabin reaction into the synthesis of CMPs, enriched the toolbox of CMPs fabrication methodology. Despite the versatility of chemical linkages and structures, which can be used for CMPs construction, the mentioned synthesis procedures all yield such CMPs as insoluble and infusible powders, due to their highly crosslinked network structure. The poor processability of CMPs is therefore a major challenge in CMP chemistry, largely preventing the "real-life" application of such materials. Consequently, recent advances have focused on the development of new protocols enabling the formation of CMPs as membranes[69], films[70, 71], aerogels[72, 73], sponges[74, 75], hydrogels[76] or coatings[77]. As example, for filtration and separation applications, CMP membranes exhibit high resistance to organic solvents/vapors owing to their all-rigid conjugated systems[78] while for adsorption applications, the conjugated structure of CMPs yield a strong affinity towards aromatic (π-π interactions)[79] and cationic molecules (cation-π interactions)[80]. These examples show that "Macroscale CMPs", *i.e.* CMPs with structural control over several length scales are of significant interest for academic and industrial research.

Macroscale CMPs should thus possess abundant, interconnected and permanent micropores within a π-conjugated, organic 3D network, combined with a defined morphology on the micrometer and even milli/centimeter length scale. To achieve such macroscale CMPs, several synthetic strategies have been suggested: Recently, post-synthetic modification (PSM) has been introduced to graft CMPs onto macroscopic objects[78 81]. However, this technique is difficult to apply to the vast majority of CMPs materials. Therefore bottom-up synthetic methods have been developed such as electropolymerization on electrodes[82, 83], synthesis of CMPs on functionalized substrates[84, 85], sol-gel reactions[86] and freeze-drying processes[87, 88] to prepare macroscale CMPs with promising properties for optoelectronics, filtration/separation and energy applications (**Figure 1**, **Outer Ring**).

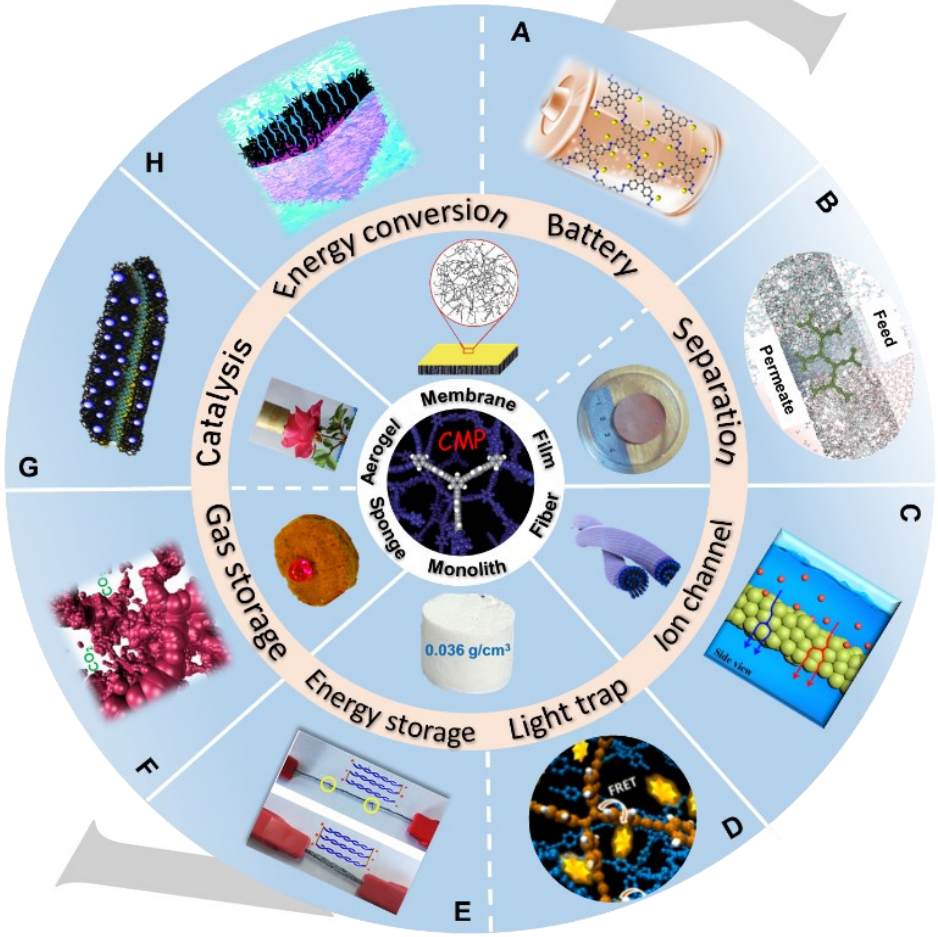

Figure 1. Schematic illustration of the classification of Macroscale Conjugated Microporous Polymers (Membrane; Film; Fiber; Monolith; Sponge; Aerogel) and their corresponding applications: (A) Battery; (B) Separation; (C) Ion channel; (D) Light trap; (E) Energy storage; (F) Gas storage; (G) Catalysis; (H) Energy conversion. (A) Reproduced with permission.[89] Copyright 2019, American Chemical Society. (B) Reproduced with

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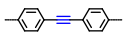
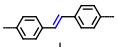
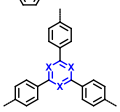
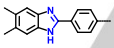
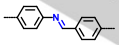
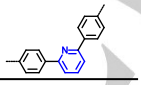
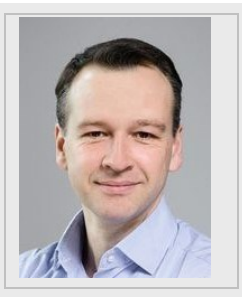
Table 1. Summary of major coupling reactions and building blocks for the preparation of Macroscale CMPs.

Coupling strategies	Monomers (C_1+C_2)		Structure
Sonagashira-Hagihara	X=Cl. Br. I. OTf		
Yamamoto	X=Cl. Br. I. OTf		
Suzuki-Miyaura	X=Cl. Br. I. OTf	Y=(OH)$_2$, (OR)$_2$, pinacol	
Heck	X=Br. I. OTf		
Cyclotrimerization	X=C. N	X=C. N	
Phenazine ring fusion			
Heterocycle linkages			
Oxidative coupling			
Schiff-base			
Alkyne metathesis			
Buchwald-Hartwig	X=Cl. Br. I. OTf		
Chichibabin reaction			

This review focuses on the preparation and application of macroscale CMPS, *i.e.* conjugated microporous polymers with controlled macroscale morphology, which enable to apply CMPs directly in devices. Functional composite materials derived from CMPs are also included. The review starts with a brief introduction on CMPs, followed by methods to prepare them as membranes, films, aerogels, sponges and miscellaneous other macroscale morphologies. In each section, applications of macroscale CMPs in gas adsorption and separation, energy storage and conversion, chemosensing, purification and filtration, heterogeneous catalysis, and others are described. Notably, other crystalline or amorphous POPs, such as COFs and PIMs, are not in the scope of this review, as macroscale morphology control of COFs/PIMs have been already reviewed [96-98]. In contrast, macroscale CMPs are still less explored but due to their structural features show another scope of possible applications.

Hongyu Zuo obtained master degree in 2020 from University of Shanghai for Science and Technology. In Master's period, she studied porous organic polymers at the Donghua University with Prof. Yaozu Liao and continued her PhD studies with him. Currently, she is working on porous organic polymer films for ionic exchange and nanofiltration applications.

Arne Thomas received his Ph.D. degree in 2004 from the Max Planck Institute of Colloids and Interfaces in Potsdam, Germany; After a post-doctoral stay at the University of California, Santa Barbara, Arne rejoined the MPIKG as a group leader. In 2009, he became a Professor at the Technische Universität Berlin where he is leading the department of Functional Materials. His research focuses on porous materials-from mesoporous inorganic materials to microporous organic frameworks.



2. Design principles of macroscale CMPs

2.1. Classification and overview

Macroscale materials refer to a class of three-dimensional materials with a size range of > 1 mm in at least one dimension. More precisely, according to the International Union of Pure and Applied Chemistry (IUPAC) definition, nano-/micro-/macroscale objects, *e.g.* particles, cover a dimensional size range of <100 nm, 100 nm ~ 1 mm, and > 1 mm, respectively. Though "macroscale materials or macroscale objects" is not a strict term defined by the IUPAC nomenclature, it has been widely employed in the field of materials science and is used here accordingly. Conventional liquid phase synthesis yield CMPs as fine powders, with particle sizes from several hundred nanometers to some micrometers, thus could not be classified as macroscale objects and are therefore not the focus of this review.

Macroscale conjugated microporous polymers (MCMPs) are CMPs with defined and designed dimensions in the macroscale (> 1 mm), commonly in the size range over 1 cm. Compared with powdery CMPs, MCMPs show a much broader scope of possible applications as they can be more easily engineered and fabricated into advanced devices, *e.g.* sensors[99, 100], smart electronics[101], or filtration membranes[70]. The possibility to create different shapes for MCMPs endow them with more functionalities, *e.g.* CMP membranes with hierarchical porosities where small pores to provide large surface areas and larger pores for efficient mass transport. Also other forms of MCMPs than thin films can be envisioned, for example, CMP sponges[74] or CMP fibers[102]. The MCMPs reviewed here are prepared by several fabrication routes

to obtain macroscopic architectures, *e.g.* top-down or bottom-up synthesis, post-synthetic modifications, and templating methods. Finally, several examples will be highlighted on how the macroscopic shape of the CMPs affects their properties and thus are decisive for the suggested application.

2.2. Synthesis and morphology control

To construct CMPs on the macroscale, the synthetic methods and reaction conditions are partially different from the conventional synthetic routes of powdery CMPs. Some major pathways can be identified, such as templating methods using hard or soft templates, or CMP formation on substrate or at interface. Other methods are post-synthetic modifications, mechanochemical or microwave-assisted methods. Each strategy demonstrates distinct advantages with respect to macroscopic architecture control, scalable production, or processing of materials' shapes, but also have limitations from a synthetic perspective. To find the right balance to control the macroscopic structural parameters (shape and size) and the microscopic porous and chemical structure is thus crucial to satisfy a specific research or application goal. **Table 2** summarizes the strengths and weaknesses of these strategies with representative references.

Table 2. Summary of major methods for the preparation of Macroscale CMPs.

	Methodology	Strengths	Weaknesses
Templating-methods	Hard templating[103]	Facile to synthesize specific morphologies on the macroscale; Obtainable hierarchical porous structures	Need to use sacrificial templates; Templates are difficult to remove; Scale-up production is restrained
	Soft templating[104]	No demoulding is required; Easily processed into macroscales structure	Formation conditions are hard to handle; Difficult to precisely control the structure
Template-free methods	Interfacial polymerization [105]	Simple reaction equipment; Mild reaction conditions; High product purity	High solvent consumption; Macroscale structures are limited by the reaction vessel; low yield
	Layer by layer polymerization [106]	Accurate control at the nanoscale; Macroscopic morphology can be adjusted;	The type of reaction is limited; Complex reaction equipment;
Post-synthetic modification[107]	–	Effective macroscale structures regulation; Adaptability to a wide range of surfaces/architectures	Constrained synthetic conditions; Quantitative analysis is difficult; Poor stability
Other methods	Microwave synthesis[108]	Shortens the reaction time while enhancing the porosity	Applicable to limited specific conditions
	Mould-method[93]	Different macroscale morphologies can be obtained simply	Poor applicability; High temperature required;

3. Preparations of macroscale CMPs

3.1. CMP films/membranes

Polymer membranes are mainly built from linear polymers and are widely used in numerous fields like separation[109], purification[110], smart interfaces[111, 112], nano-devices[113] or biomedical applications[114]. Recently, conjugated microporous polymer membranes (CMPMs), have been developed and attracted much interest since they combine the benefits from both, conventional polymer membranes (good flexibility, light-weight, portability, efficient mass transport) and CMPs (intrinsic meso-/microporosity, (semi-)conductivity, π-conjugation, structural rigidity, *etc.*). In 2012, Cooper *et al.* reported a first example of soluble conjugated microporous polymer which can be processed into thin films *via* simple drop-casting[50]. Thereafter, several other possibilities to prepare thin films or membranes of CMPs were reported.

3.1.1. Electropolymerization

Electropolymerization (EP) is an electrochemical process to fabricate (crosslinked) polymer films from molecular or polymeric precursors on electrodes[115, 116]. This technique most often applies oxidative polymerization protocols with particular monomers[117], *e.g.* thiophene, pyrrole or carbazole-functionalized ones[30, 118, 119], and can be applied to create CMP membranes/films [29] with defined structure, desired thickness and surface orientation on different conductive substrates (**Figure 2a**). For instance, Sezer *et al.*[120] reported the formation of crosslinked polymer films by electropolymerization of bis(phenylthieno[3,2-b]thiophenes)–(4,4'-dinonyl-2,2'-bithiazole) (PhTT-NBT-PhTT) co-monomer with trifluroborate–diethyl ether (BF_3·DEE). Nicholls *et al.* [121] described an electropolymerized polymer film composed of four different monomers for recognition devices.

Ma *et al.*[124, 126] and Jiang *et al.*[127-129] reported a series of pioneering works utilizing electropolymerization to fabricate CMP films (Figure 2a, b). Initially, Ma *et al.* reported CMP films from TPTCz (tetrakis(4-(9H-carbazol-9-yl)phenyl)methane), that is from tetrahedral tetraphenylmethane cores functionalized with peripheral carbazole moieties. The formed films presented uniform micropores with diameters of less than 1 nm as seen from HR-TEM (high resolution transmission electron microscopy) (Figure 2c). Later, the same group designed a novel 1,2-bis(4-(9'H-[3,6-tercarbazol]-9'-yl)phenyl)diazene (DTCzAzo) monomer to synthesize CMP films[123] with homogeneous micropores and large specific surface areas (**Figure 2a, d**). As this yield a high

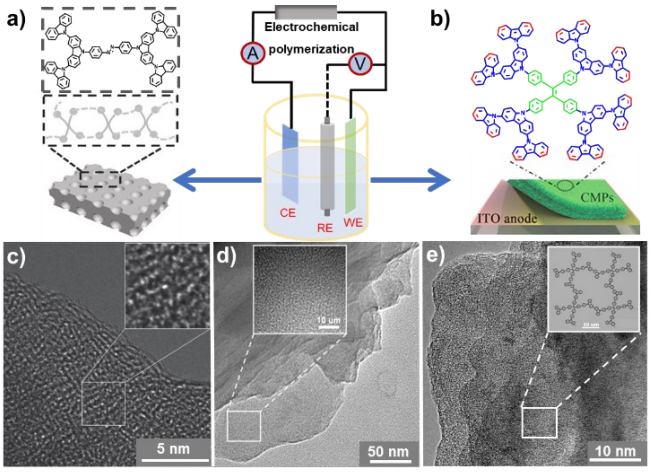


Figure 2. (a) The fabrication of CMP films by electropolymerization (EP) in the condition of DTCzAzo; (b) Chemical structure of TPETCz and the fabrication of CMP films by ECP; TEM image of (c) TPTCz EP film, (d) DTCzAzo EP film and (e) polyTPECz film. (a) Reproduced with permission.[122] Copyright 2020, Wiley-VCH. (b) Reproduced with permission.[123] Copyright 2017, Wiley-VCH. (c) Reproduced with permission.[124] Copyright 2013, Wiley-VCH. (d) Reproduced with permission.[123] Copyright 2017, Wiley-VCH. (e) Reproduced with permission.[125] Copyright 2015, Wiley-VCH.

work function anode interlayer, the doped CMP film showed a significant power conversion efficiency of 7.56% in polymer solar cell and a luminous efficiency of 20.7 cd A^{-1}, which demonstrates its great potential as an electrode interlayer in organic electronics. Simultaneously, Jiang and co-workers[125] successfully electropolymerized 1,1,2,2-tetrakis(4'-(9H-carbazol-9-yl)-[1,1'-biphenyl]-4-yl)ethene (TPECz) into aggregation-induced emission (AIE)-based highly emissive CMP films. The formed TPECz-CMP films were investigated with a high-resolution transmission electron microscope and found to be highly microporous (**Figure 2e**). Such CMP films exhibited high rate of photoinduced electron transfer, which renders the films able to detect explosives (at low parts-per-million concentrations) selectively and sensitively. As electropolymerization is a facile and general applicable technique, it has been applied frequently to construct CMP films/membranes in the following years. Liu *et al.*[122] reported a highly porous CMP film constructed from dendrimer 1,1,2,2-tetrakis(4-(9'H-[9,3':6',9''-tercarbazol]-9'-yl) phenyl)ethane (TPETCz) (**Figure 2b**). The CMP films with high Brunauer–Emmet–Teller (BET) surface areas (1042.5 m^2 g^{-1}) was obtained by the nonlocal density functional theory (NL-DFT) method. Proved to be highly luminescent and thus could be used as a 2D fluorescence sensor thereafter. Recently, Tang and Li *et al.*[107, 130] electropolymerized a 1,3,5-tri(2-thienyl)-benzene (TTB) containing CMP membrane. An ultra-fast organic solvent permeability (32 L m^{-2} h^{-1} bar^{-1}) was achieved for such CMP membranes. After fine tuning of the pore size within 1 nm by post-oxidation of the thiophene portion of CMPs, the organic solvents nanofiltration (OSN) permeability of methanol was still as high as 21 L m^{-2} h^{-1} bar^{-1}, and the molecular weight cut off (MWCO) was greatly reduced from 800 g mol^{-1} (before modification) to 500 g mol^{-1}, which made them promising candidates for various fluid separations. Later, Thomas *et al.*[131] synthesized a dithienothiophene(DTT)-based CMP film through oxidative polymerization and electrochemical deposition on an ITO electrode. A complete reversible color change from orange-red (0 V) to light green (1.2 V) and finally light blue (1.4 V) was monitored by spectroelectrochemical measurements during the redox process. Such DTT-based CMP film with reversible elecrochromic effects can be treated as promising materials for differed applications, like detection of ammonia-based explosives, chemosensing, or working as smart windows.

3.1.2. Templating method

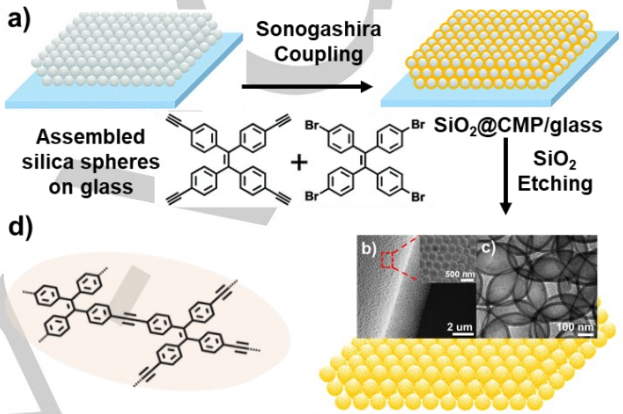


Figure 3. (a) Synthetic schemes for MA-MON-F; (b) SEM images of MA-CMP-F; (c) TEM images of MA-CMP-F; (d) Structure of CMP-F. (a-d) Reproduced with permission.[132] Copyright 2018, Royal Society of Chemistry.

The morphological structure of CMP films can also be modified by applying templates. For instance, Son *et al.* applied silica spheres (with an average diameter of 400 nm) as templates for the synthesis of CMP membranes on glass plates. A Sonogashira coupling reaction of tetramethylethylphenylethylene with tetramethyl-bromophenylethylene was applied to create CMPs surrounding the surfaces of the silica spheres. After removal of the silica spheres, a novel CMP film with hollow structures (Figure 3) was obtained. The formed CMP film illustrated ordered inverse opal structures[132]. Later, Liu and Yin *et al.*[133] prepared a 2D CMP (4,4'-di(9H-carbazol-9-yl)-1,10-biphenyl as monomer) film on rGO / SiO_2 /Si substrates *via* photomask-assisted solid-state photopolymerization under ambient conditions. Several geometries (*e.g.* circular, disc-shaped, square, oblong and irregular) were obtained by applying different photomasks and regulating polymerization durations. The patterned 2D CMP/rGO heterostructure was used as an organic vertical field-effect transistor, which exhibited typical p-type behavior with a high on/off ratio of up to 2.0 X 10^4. Subsequently, Ryu *et al.* [134] used tetra(4-ethynylphenyl)ethylene and 1,4-diiodobenzene as building blocks to synthesize CMP films as aligned tubes in the cylindrical pores of anodized aluminum oxide (AAO) plates through Sonogashira coupling. Such a CMP film was applied for nitrotoluenes sensing upon the quenching of the AIE (aggregation-induced emission) phenomenon. Alternatively, except polymerizing based on substrates, researchers studied the

pre-functionalization of the substrates by monomers containing active groups, such as –OH, –NH, –SH, bromobenzene, and so on, for the growth of 2D CMP films. For instance, Thomas *et al*.[135] synthesized microporous polymer network (MPN) films by functionalizing gold substrate with 4-bromothiophenol *via* rinsing and drying process, followed by Yamamoto coupling polymerization reactions. Physical and chemical properties of such MPN films were thoroughly studied. Lee *et al*.[104] applied porous polyacrylonitrile (P-PAN) as a robust supporting matrix for synthesizing CMP-BT films from monomers 4,7-Dibromobenzo[c]-1,2,5-thiadiazole and 1,3,5-phenyltriboronic acid, with the tris(pinacol) ester embedded into P-PAN. Thereafter, electrospinning was applied to obtain CMP@P-PAN nanofiber films. These films were used as photocatalyst for degrading rhodamine B. In addition, CMP@P-PAN can also catalyze the oxidative coupling reaction of benzylamine derivatives in organic synthesis under visible light irradiation.

3.1.3. Interfacial polymerization method

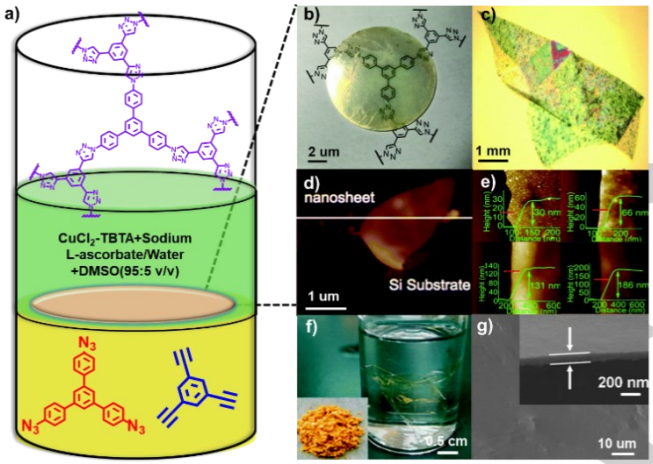


Figure 4. (a) Schematic illustration of the liquid/liquid interfacial synthesis of CMP-NF1. (b) Photograph on a glass substrate with a diameter of 12 cm. (c) Optical microscopic image on a glass substrate. (d) AFM height image on a HMDS-modified SiO_2/Si (111) substrate; (e) Tapping-mode AFM images of CMP-1-nanofilms synthesized using different monomer concentrations (a, TEB: 0.0875 mM, DIB: 0.125 mM; b, TEB: 0.175 mM, DIB: 0.25 mM; c, TEB: 0.263 mM, DIB: 0.375 mM; d, TEB: 0.35 mM, DIB: 0.5 mM), and their height. (f) Photo image of the CMP-1 nanofilm (in the ethyl acetate phase), inset: CMP-1 bulk powder; (g) SEM image of CMP-1. (a-d) Reproduced with permission. [137] Copyright 2020, Royal Society of Chemistry. (e-g) Reproduced with permission.[105] Copyright 2017, Royal Society of Chemistry.

Interfacial polymerization[136] is a synthetic strategy commonly used in membrane formations. When two active monomers are dissolved separately in incompatible solvents, *e.g.* water and nonpolar organic solvents, the polymerization will be initiated at the solvent interface and form a dense membrane afterwards. It is worth to mention that the nature of the monomers will directly affect the reaction rates and the properties of interfacial polymerized membranes. For instance, Sakamoto and coworkers[137] initiated the cycloaddition reaction of alkyne- and azide-functionalized monomers at a water/dichloromethane interface and obtained 1,2,3-triazole-linked CMPMs (termed as CMP-NF1) with high aspect ratio and good thermal/pH stability (**Figure 4a**). The formed luminescent CMPMs can be easily transferred to different substrates and exhibit good mechanical properties (**Figure 4b**). Moreover, the CMP-NF1 exhibited a sheet-like morphology with a folded structure (**Figure 4c**). Atomic force microscopy showed an NF1 domain partially folded over, again indicating that it was a sheet (**Figure 4c**). Apart from, the diffusion rate of the monomer through the forming membrane at the interface is a key factor affecting their thickness. Since interfacial polymerized CMPs are generally soft, substrates are often added as supporting materials in the preparation process. In order to obtain high-performing filtration membranes, Huang and Tang *et al.* [138] synthesized CMP membranes with thicknesses of less than 10 nm through the interfacial polymerization of ethidium bromide (EtBr) with benzoyl chloride (TMC). The ultrathin film was transferred to an ITO substrate, which makes the films retain their integrity in a large scale. Then the as-synthesized membrane was firstly transferred onto a polyacrylonitrile (PAN) substrate for nanofiltration measurement. Such CMPMs exhibited high water permeability (40 L m^{-2} h^{-1} bar^{-1}) and good congo red dye repellency in organic solvents nanofiltration. Thus, molecular separation was realized. Similarly, Wu et *al*.[105] successfully prepared CMP nanomembranes at the water/organic interface by Sonogashira-Hagihara and Schiff base reaction (**Figure 4e, f, g**), for which the thicknesses can be restrained within 30-200 nm *via* alternating monomer concentrations (**Figure 4e**). These works revealed a facile route to prepare functional 2D porous polymer nanomembranes with large domain sizes.

3.1.4. Layer-by-layer method

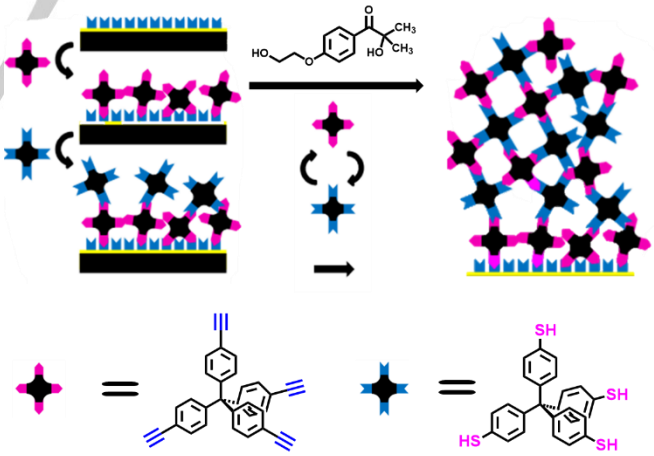


Figure 5. LbL synthesis with TPM-SH and TPM-alkyne using light-induced thiol-yne click reaction reaction in the presence of the photoinitiator 2-hydroxy-1-[4-(2-hydroxyethoxy)phenyl]-2-methylpropan-1-one. Reproduced under the terms of the CC-BY Creative Commons Attribution 4.0 International license (https://creativecommons.org/licenses/by/4.0).[141]

Layer-by-layer (LbL) assembly[139] is a thin film fabrication process in which building blocks with complementary interactions are deposited onto substrates *via* alternating sorption. It is a

conceptually simple, practically powerful technique to produce thickness-precise films at nanometer scale. Most LbL films are dense and well-suited for molecular separation and barrier films applications. By modulating post-treatment and assembly conditions, LbL films were made porous for applications like sensors, smart coatings, and controlled delivery vehicles. Based on the successful preparation of metal organic framework (MOF) layers[140], Tsotsalas *et al.*[106] reported a defect-free CMPM synthesized from tetra (4-azidophenyl)-methane and tetra (4-ethynylphenyl)-methane *via* layer-by-layer assembly, and applied these films for gas selectivity and ferrocene interception testing. Later, the same group applied two building blocks tetrakis(4-sulfanylphenyl)methane (TPM-SH) and tetrakis(4-ethynylphenyl)methane (TPM-alkyne) to grow CMP films *via* light induced solid-liquid interfacial LbL thiol-yne click reaction[141] (**Figure 5**). The resulting nanomembranes exhibited high mechanical stability and can be transferred to arbitrary supports, which endowed such CMP films wide applicability in organic electronics or gas/liquid phase separation devices. Followingly, a further report from the same group[142] proposed a strategy to synthesize functionalized CMP nano-films based on two parallel click chemistries, copper(I)-catalyzed alkyne-azide cycloaddition (CuAAC) and light induced thiolyne reaction. This method was fast in response, produced no by-products and allowed direct light-patterning of the film surface, which exhibited additional possibilities for photo-patterning of chemical mechanical polishing materials.

3.2. CMP aerogels, sponges and monoliths

3.2.1. CMP aerogels

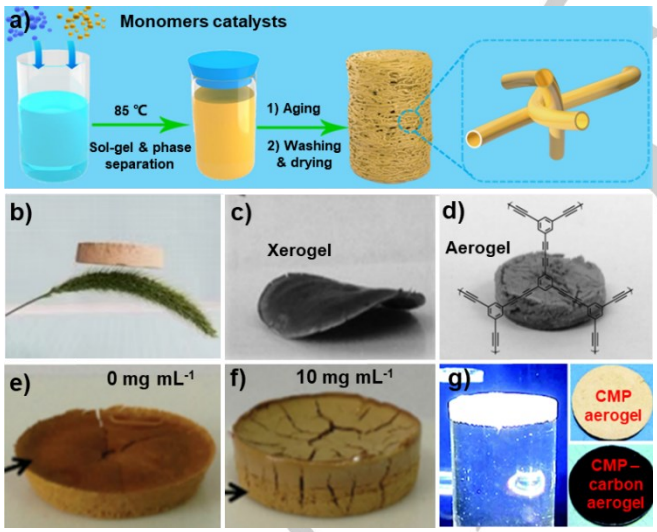


Figure 6. (a) Synthesis schematic diagram of the PCMPAs; (b) Photograph of CMPCA-1 on a dog tail grass. (c,d) PTEB aerogels acquired from corresponding xerogel and aerogel *via* freeze-drying. (e, f) corresponding aerogels obtained after freeze-drying with 0/10 mg mL^{-1} CTAB; (g) Left: photograph of CMP aerogel under 2 kW m^{-2} illumination for 5 min. Upper right: photograph of CMP aerogel of 3.2 cm in diameter and 2 mm in height, Lower right: CMP aerogel treated with carbon black. (a, b) Reproduced with permission.[145] Copyright 2020, Elsevier. (c,d) Reproduced with permission.[150] Copyright 2014, Wiley-VCH. (e, f) Reproduced with permission.[151] Copyright 2015, The Authors, published by Wiley-VCH. (g) Reproduced with permission.[95] Copyright 2018, Royal Society of Chemistry.

Porous polymer aerogels are hierarchical porous materials with three-dimensional network structures[143]. Owing to their rich porosities, aerogels feature the characteristics of ultra-low density, light-weight, high specific surface area and low thermal conductivity[144]. In recent years, conjugated microporous polymer aerogels (CMPAs) have been frequently used in a variety of fields, including energy storage, catalysis or energy conversion. Aerogels are normally prepared *via* sol–gel chemistry and followed by supercritical drying or freeze-drying. The "sol-gel" method is normally connected to inorganic materials (*e.g.* silica, carbon, and transition metal oxides) to create films, particles, xero- or aerogels. However, this term can be as well applied to CMP synthesis when in the first step the formation of homogeneous dispersion of small polymer oligomers or particles are observed which subsequently grow to a gel network. If these steps are followed by a convenient drying process, CMP aerogels can be obtained as well. The successful formation of CMP aerogels and also CMP sponges described below were determined by several crucial factors: (1) Solubilities of reacting monomers: outstanding solubilities can promote the formation of homogeneous solutions of reacting monomers and catalysts [145]; (2) activities and molar ratios of the reacting monomers: it can accelerate the phase transition from solution to gel during reaction process[146]; (3) micro-morpholgies of the resultant CMPs: nanotubular or nanofibrous structure can make the CMPs intertwined to form a 3D macro-structure[147]. Zhu *et al.* [145] reported a CMP aerogel constructed from 2-amino-3,5-dibromopyridine and 1,3,5-triethynylbenzene (TEB) by the sol-gel method (**Figure 6a**). By adjusting the molar ratio of the monomers, a high mechanical strength and light-weight CMPA with macroporous structure constructed from interconnected nanotubes was successfully prepared (**Figure 6b**). The CMPA was fabricated into a double-layer solar steam generator upon spraying a thin layer of polypyrrole on its surface. Solar steam generation reached a high energy conversion efficiency of 80% at power density of 1 kW m^{-2}. The process conditions such as monomer molar ratio, solvent volume, sol-gel reaction temperature and reaction time, and the drying process directly determined the microstructural morphology and performance of the aerogels[148, 149].

Du *et al.*[150] firstly reported high-performance CMP aerogels derived from a drying process. Utilizing poly(1,3,5-triethynylbenzene) (PTEB) as the precursor, PTEB-CMP xerogels/aerogels (**Figure 6c, d**) with a Brunauer-Emmett-Teller (BET) surface area of 1085/1701 cm^2 g^{-1} were obtained *via* vacuum/freeze-drying process. It is worth mentioning that the specific surface area (SSA) of this aerogel is ~200% higher than the reported powdery ones (842-955 m^2 g^{-1}) [74, 152]. Subsequently, the same group[151] prepared a homogeneous, hierarchical porous CMP-based aerogel assisted by fluorosurfactants, using the Glaser coupling reaction of 1,3,5-triethynylbenzene followed by a freeze-drying process (**Figure 6e, f**). Such hierarchical porous aerogels exhibited tunable and high gas sorption capacity of CO_2

and CH_4 (0.41-3.47 mmol g^{-1} for CO_2 and 0.12-0.95 mmol g^{-1} for CH_4) but also exceptional oil cleanup performance (20-48 times weight gain) among reported microporous organic polymers[153, 154]. Subsequently, as the first CMP example for photothermal conversion, Mu *et al.* [95] prepared a CMP aerogel by sol-gel synthesis through Sonogashira-Hagihara reaction with 1,3,5-triethynylbenzene, 1,4-dibromobenzene and 4,4'-dibromobiphenyl as monomers. After loading carbon black on the surface, the formed CMP-carbon aerogels can effectively generate steam under solar irradiation (**Figure 6g**).

3.2.2. CMP sponges/monoliths

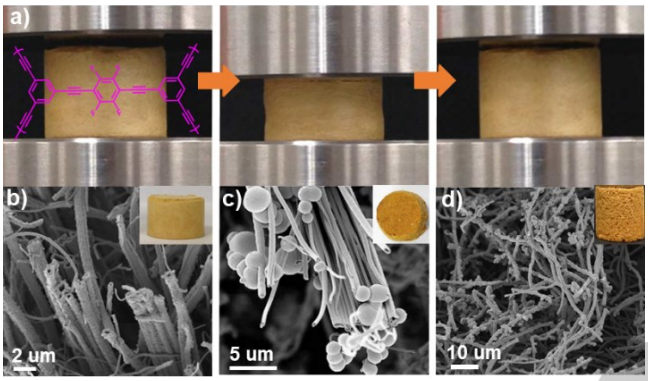


Figure 7. (a) Chemical structure of a conjugated microporous polymer sponge and photographs under compression and release conditions (MPs); SEM images of (b) the MOP sponge shown above, (c) MP-0.8, and (d) PVASi@TEDB-NH2. (a, b) Reproduced with permission.[155] Copyright 2017, American Chemical Society. (c) Reproduced with permission.[156] Copyright 2015, The Author(s), published by Springer Nature. (d) Reproduced with permission.[157] Copyright 2016, Elsevier.

A sponge is originally a primitive invertebrate multicellular organism. During the development of materials science, the term "sponge" has been gradually extended to porous elastic materials [158]. Alternatively, the main types of sponges include oxide,[159] carbon-based,[160] MOF,[161] or polymer sponges,[162] Initially, bottom-up methods were attempted to synthesize CMP-type sponges. Nevertheless, CMPs can be mold into gels or monolithic polymers that were visible to the naked eyes, but they are fragile after drying process. Similar to CMP sponges, CMP monoliths refer to microporous polymeric materials with 3D continuous, interconnected porous structures in a single piece. Above all, CMP sponges and monoliths can be differentiated from CMP aerogels because neither of them featured the characteristics of ultra-low density and light-weight. However, CMP sponges and monoliths are difficult to be distinguished practically. Hence, we termed such materials as "CMP sponges" uniformly in this article. The design and construction of CMPs into a sponge with good mechanical strength combines benefits from both sides. The most common preparation methods of CMP sponges include: (1) one-pot "sol-gel" type self-assembly fabrication method using multifunctional building blocks or (2) synthesis on a sponge skeleton or surface for CMP sponge composite materials. Li *et al.*[74] reported a super-hydrophobic conjugated microporous polymer (HCMP) *via* Pd(II)/Cu(I)-catalyzed homocoupling polymerization of 1,3,5-triethynylbenzene. By loading such CMPs (7.0 mg cm^{-3}) on the skeleton of synthetic sponges, the hydrophilic sponge can be changed to lipophilic, and the absorption of p-nitrobenzene reached 3300 wt%. Furthermore, this CMP sponge maintained excellent adsorption properties for oils and organic solvents. Chang *et al.*[155] reported a single-piece CMP sponge, which was synthesized by Sonogashira–Hagihara cross-coupling reaction from 1,4-dibromo-2,5-difluorobenzene and 1,3,5-triethynylbenzene through sol-gel process. This sponge exhibited a nanotube-like structure and a high mechanical stability against compressive forces (**Figure 7a, b**). By thiol−yne modification the MOP sponge can be functionalized with thiol groups and subsequent reduction of Ag^+ ions on the surface yield silver nanoparticles on its surface. The final CMP sponge-Ag can be used for the reduction of 4-nitrophenol in aqueous solution and recycled after reaction. The same group[156] used also thiophene-based monomers to synthesize a tubular-CMP sponge (MP-0.8) through a stepwise reaction and the structured entanglements of the resultant nanotubes. Tubular polymer fibers with a diameter of hundreds of nanometers together with spherical particles on the micrometer scale can be observed with complex entanglements (**Figure 7c**), exhibiting hierarchical pores with specific surface areas up to 512 m^2 g^{-1}. Kim *et al.*[157] prepared a core-shell structure compressible CMP composite sponge using electrospun nanofibers (termed as PVASi) obtained from a solution of PVA and TEOS as the substrate for the polymerization reaction. The CMP was synthesized by Sonogashira–Hagihara coupling of 2,5- dibromoaniline (TEDB-NH_2) and 1,3,5-triethynylbenzene in a dispersion of homogenized PVASi nanofibers (**Figure 7d**). It was suggested, that the amino groups of the monomer and growing polymer network can form hydrogen bonds with the hydroxyl groups on the surface of PVASi, hence the polymer grew uniformly on the surface of the nanofibers.

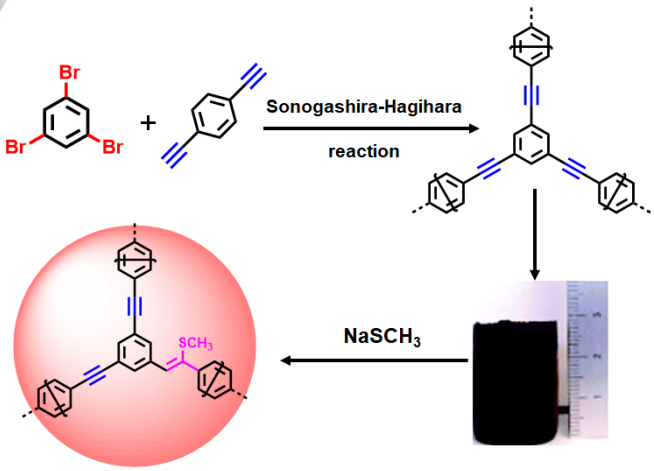


Figure 8. Synthetic scheme for the conjugated microporous polymer monolith (CMP-1 and CMP-S). Reproduced with permission.[163] Copyright 2015, Royal Society of Chemistry.

Later, Vilela and Xu *et al.*[163] synthesized a CMP sponge *via* a copper-free Sonogashira–Hagihara cross-coupling reaction in water under aerobic conditions and without stirring and directed by a "sol-gel" mechanism of the aromatic building-blocks during synthesis (**Figure 8**). Post-modification of the polymer with thiolyne chemistry followed by Ag uptake led to a catalytically active material for a hetero-cyclization reaction. Subsequently, Chang *et al.*[164] synthesized a monolithic, palladium acetylide-based conjugated microporous polymer (Pd-CMP) by using triethylammonium chloride (TEACl) with as the template. Since TEACl are insoluble rod-like crystals, they were applied as hard templates in CMP polymerization growth followed by removal of them *via* pyrolysis at 300/500 °C. Similarly, Kim *et al.*[165] reported a bottom-up synthetic method for the preparation of a compressible microporous monolithic material with a sponge-like structure. This CMP sponge was composed of 1,3,5-triethynylbenzene (TEB) and benzene or thiophene as the backbone. After incorporated with carbon additives during condensation polymerization, they were capable of regulating macroscopic conductivity in response to external compressive force.

3.3. Macroscale CMP Composites

The macroscale CMPs discussed above are in most case entirely composed of the CMP itself forming either 2D thin films or free-standing membranes or 3D monoliths. However, the shape of macroscale CMP can be also determined by a pre-formed substrate on which the CMP is coated or blended. Beside controlling the macroscale morphology, this approach has also the advantage of combining the useful properties of CMPs, as described above, with beneficial properties of the substrate, *e.g.* flexibility or electric conductivity.

CMP fibers. CMPs with fiber-like morphologies are interesting to obtain high performance flexible macrostructures which can be even further processed into fabrics. There have been few reports in recent years on fiber-shape CMPs. For instance, Yuan *et al.* [166] fabricated fluorescent, nanofibrous networks based on light-emitting CMPs by electrospinning of CMP/poly(lactic acid) (PLA) mixtures. The resulting nanofiber film exhibits high flexibility, porosity and a high surface area to volume ratio. The CMP-based nanofiber film was used as a sensor for detecting nitroaromatic and benzoquinone vapor and oxidized metal ions. However, the fabrication of macroscale fibers with CMPs had not been achieved yet. Recently, Liao *et al.* [81] reported a series of CMPs grafted on carbon nanotube fibers (CNF@CMPs) *via* a Buchwald−Hartwig coupling reaction between tris(4-bromophenyl)amine as a core and various amines as linkers (**Figure 9a**), which examplified the possibility to create macroscale CMP-based fibers for weaving and fabric formation. The resulting CNF@CMPs exhibited excellent electrical energy storage capacity and were successfully applied in fabricating high-performance wearable supercapacitors.

CMP meshes and papers. In 2017, Sun *et al.*[167] reported two types of conjugated microporous polymers coated superhydrophobic devices. They synthesized one superhydrophobic CMP (UPC-CMP-2) by using iron(III)-5,10,15,20-tetrakis-(4-bromophenyl)porphyrin and 1,3-diethynylbenzene as building unit *via* Sonogashira-Hagihara coupling reaction. The UPC-CMP-2 exhibited unique super-hydrophobicity for which the water contact angle was 158° (**Figure 9b**). It was proven that UPC-CMP-2 can be coated onto a steel mesh and sponge easily. The coated devices showed excellent performance in oil/water separation and water purification, and the coated mesh demonstrated diesel/water and crude oil/water separation behaviors with high efficiencies being 99.3% and 99.8%, respectively. Furthermore, Li *et al.*[168] synthesized one fluorine-rich conjugated microporous polymer (FCMP) *via* Sonogashira-Hagihara coupling reaction. This material combined advantages both from the abundant hierarchical porosity (1194 m^2 g^{-1}) and super-hydrophobicity (water contact angle of 160°). Ultimately, they coated FCMPs on printing paper to directly observe the hydrophobic properties of the resulting paper sheets (**Figure 9c**).

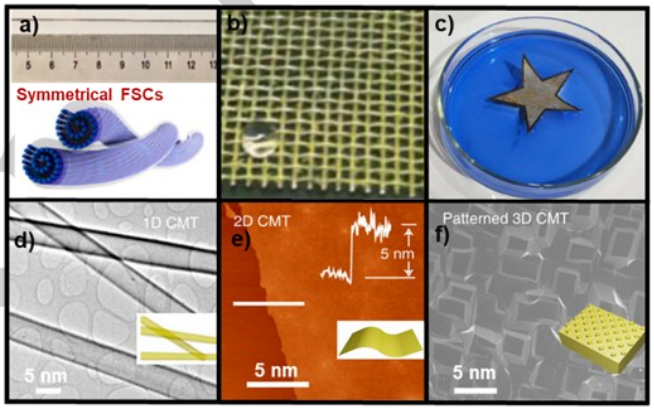


Figure 9. (a) Typical CNF@CMP fibers with an average diameter of ~200 μm and schematic illustration of the symmetrical fiber-shaped supercapacitors; (b) Photographs of a water droplet on the stainless steel mesh with UPC-CMP-2 coating; (c) snapshots of a piece of as-treated superhydrophobic print paper immersed into water and dyed with pigments; (d) TEM image showing the surface of a patterned 3D CMT; (e) AFM topography image of ~5 nm-thick CMT film on Si wafer; Insert: corresponding height profile. (f) SEM image of 1D CMT nanotubes. (a) Reproduced with permission.[81] Copyright 2020, American Chemical Society. (b) Reproduced with permission.[167] Copyright 2017, Elsevier. (c) Reproduced with permission.[168] Copyright 2018, Elsevier. (d-f) Reproduced with permission.[93] Copyright 2020, The Author(s), published by Springer Nature.

Thermoset CMPs. Loh *et al.*[93] developed a soluble and re-processable conjugated microporous thermosetting polymer (CMT) polymerized from the precursor 3,6,12,15-tetrabromotetrabenzo[a,c,h,j]phenazine (3-TBTBP). Different substrates (a holder with patterned bottom plate for 3D CMT, Silicon wafer for 2D CMT, Cu nanowires for 1D CMT) were used as templates on which the monomer could be coated by sublimation and melting. Polymerization *via* thermal-initiated debromination and concomitant C–C cross-coupling reaction then yielded 3D, 2D, and 1D CMTs (**Figure 9d, e, f**). The as-formed CMTs maintained the processability of plastic thermosets, and inherited the rigidity and porosity from CMPs as well. Moreover, a large-area ultrafiltration membrane with a controllable thickness were prepared *via* filtration of CMT ultrathin sheets dispersible in chloroform. The resulting porous membrane was utilized as a two-

dimensional molecular sieve membrane with a selectivity to H_2 over CO_2, O_2, N_2, CH_4, C_3H_6 and C_3H_8. As a hydrogen separation membrane, it exhibited broad prospects in numerous fields, *e.g.* industrial scaling-up fabrication and commercial hydrogen separation.

4. Functional exploration and applications

As mentioned above, macroscale conjugated microporous polymers allow structural control over multiple length scales. Such a control on chemical structure, porosity on the nanometer scale to macroscale morphologies can largely expand their areas of applications, especially in field where fine powders are impractical. For instance, advanced applications like gas separations, nanofiltration, in optoelectronic devices or sensing materials require CMPs in forms of films, membranes, monoliths or others. In this part, we summarize the most recent applications realized by macroscale CMPs, including gas adsorption/separation, energy storage, energy conversion, light-emitting/chemosensing, filtration and purifications, and organic catalysis.

4.1. Gas adsorption and separation

4.1.1. Gas adsorption

The adsorption and separation of gases is an often reported application of CMPs and also macroscale CMPs, even though a conjugated π-system is not primarily required to adsorb large amount of gases in the framework. However, CMPs provide robust frameworks with small pores and very high surface areas, thus all these properties make them valuable for these applications. Besides, even though not discussed commonly in the literature, the conjugated backbone structure might influence gas adsorption enthalpies in comparison to aliphatic systems. Gas/liquid adsorption can be readily realized by powdery CMPs, and showed good performance in numerous examples[169, 170]. However, in large scale adsorption installations, macroscale CMPs, *e.g.* CMP aerogels or sponges, are easier applicable and recyclable and therefore received rising research interest. For instance, Du *et al.* [143] reported a large surface area CMP aerogel with an enhanced CO_2 adsorption capacity (up to 3.47 mmol g^{-1}) over previously reported conventional CMP CO_2 adsorbent. (less than 2.5 mmol g^{-1})[171, 172]. Such monolithic CMP aerogels were obtained by Glaser-coupling reactions of 1,3,5-triethynylbenzene under an air atmosphere and followed byt freeze-drying. Subsequently, the same group[173] synthesized similar porous poly(1,3,5-triethynylbenzene) aerogels, with a hierarchical porous structure and strong hydrophobicity induced by fluorinated surfactants. High gas adsorption capacity (0.41-3.47 mmol g^{-1} for $CO_2$and 0.95 mmol g^{-1} for CH_4) and excellent oil pollution treatment capability were achieved, with an oil/water separation rate of nearly 95% [78].

4.1.2. Gas separation

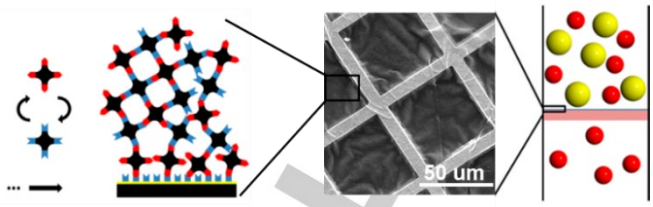


Figure 10. Schematic representation of the layer-by-layer synthesis of a CMP on functionalized surfaces and gas separation. Reproduced with permission.[174] Copyright 2014, American Chemical Society.

Owing to their small pore size and large specific surface area, CMPs have been successfully utilized in the field of gas adsorptions. However, their powdery forms impose restrictions on the gas separation utilizations because gases can flow through the interstitial voids of the powders, which strongly reduce their separation capacity. Consequently, membranes or thin films of CMPs have to be fabricated for gas separation applications. Lindemann *et al.*[174] reported defect-free CMP nanomembranes manufactured through layer-by-layer synthesis on a substrate (**Figure 10**). By chosen two differently functionalized monomers, one with an azide, the other with alkyne groups, a freestanding thin CMP film can be fabricated by the "layer-by-layer" approach *via* click coupling. Permeation experiments on the films were tested with 8 different gases including He, H_2, CO_2, Ar, O_2, N_2, CH_4, and C_2H_6, and high selectivity values between these gases were proven. The CMP-membrane showed a clear molecular sieving effect favorable, in terms of gas transport through the membrane, for the gases with smaller kinetic diameter (He and H_2 and, to a lesser extent, CO_2). The selectivity of this membrane was ~36 for H_2/N_2 and ~6 for O_2/N_2, with a permeability of ~4 Barrer for H_2 and ~0.7 Barrer for O_2.. Shortly after, Wang *et al.* [175] developed a CMP membrane prepared by space-confined polymerization of monomers in the mesoscopic channels of a water ultrafiltration polysulfone (PSU) membrane. This membrane demonstrated aging-resistant behaviors with size-sieving ability for gas molecules through intrinsic ultra-micropores. The as-synthesized CMP@PSU membranes exhibited good selectivity of H_2 over CO_2 (79.4), N_2 (89.5), CO (107.0) and CH_4(126.3), with a high H_2 permeability of 4924 Barrer. This membrane also acquired good aging resistance of only 2.6% decrease after 180 days. These works illustrated the broad prospects of CMP membranes/films in the field of large-scale gas separations.

4.2. Energy storage and conversion

4.2.1. Supercapacitors

Supercapacitors are energy storage devices that are exploited for high-performing power supplies of electrical cars or miniatured electronic products. MCMPs can show effective energy-storing behaviour because of the complementary combination of π-conjugation, rich porosities and macroscopic structures. Ma *et al.* [179] reported a CMP film-based supercapacitor obtained by electropolymerization. With zinc porphyrin as the core, this highly crosslinked CMP film (denoted as Zn-mTCPP) exhibited high stability at high current density, with a specific capacitance value

of 142 F g^{-1}. Later, Zhou and co-workers[76] integrated CMPs onto macroporous graphene aerogels with three-dimensional hierarchical pore structures. When such a CMP-graphene composite aerogel was applied as the supercapacitor electrodes, it presented a high capacitance of 208 F g^{-1} at 20 mA g^{-1}. Nevertheless, the capacitive performances of CMPs are still restrained by their low conductivities. To tackle this challenge, Liao *et al.*[176] introduced a redox-active conjugated microporous polyaminoanthraquinone (PAQ) network *via* a one-step Buchwald-Hartwig coupling. Owing to their good dispersibility in polar solvents, the PAQ-type CMPs can be uniformly coated onto carbon papers to obtain macroscale flexible electrodes. Such CMPs-based electrodes, consisting of anthraquinonylamine and triphenylamine (PAQTA) exhibited outstanding electrochemical energy storage performance (576 F g^{-1} at 1.0 A g^{-1}) owing to their hierarchical porosity and high redox activity. When applied as electrodes in asymmetric two- electrode supercapacitors, a high energy density (60 Wh kg^{-1}) and Coulombic efficiency (up to 97%) was achieved (**Figure 11a, b).** Similarly, CMP-carbon composites were applied for supercapacitive energy storage thereafter.[180]

4.2.2. Batteries

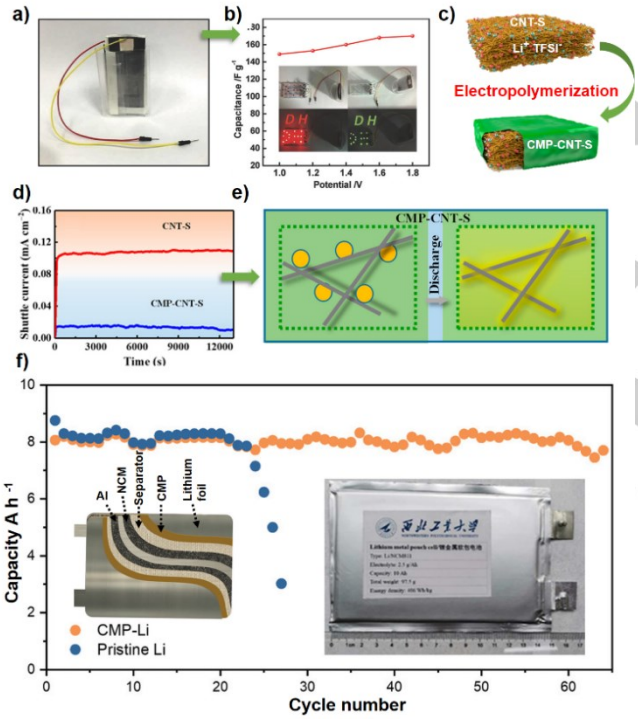


Figure 11. (a) Photograph of the ASC device. (b) Specific capacitance as a function of the various potential windows with the inset showing a PAQTA//AC ASC device synchronously powers up 14 red or green LED bubbles assembled in a DH symbol (Donghua abbreviations). (c) CMP-CNT-S cathode prepared *via* the electropolymerization sealing process. (d) Shuttling currents profiles of the cells. (e) Schematic illustration of the evolution of CMP-CNT-S electrodes during the discharge process. (f) Cycling performance of 10 Ah pouch cell with CMP-Li anode and pristine Li-anode. Inset (left): Illustration of the single-layer unit composed of NCM811 cathode, separator, and CMP-Li. Al foil was used as cathode current collector and Li-foil was directly used as anode current collector. Inset (right): photograph of 10 Ah pouch cell. (a, b) Reproduced with permission.[176] Copyright 2018, Wiley-VCH. (c-e) Reproduced with permission.[177] Copyright 2020, Wiley-VCH. (f) Reproduced with permission.[178] Copyright 2020, American Chemical Society.

Macroscale CMPs are also applied directly in batteries as anode/cathode materials or protective layers. For instance, Zhang *et al.*[181] synthesized a highly porous (657 m^2 g^{-1}) CMP film using 4, 4', 6, 6'-(4-triphenyl)-2, 2'-bitriazine (TPADTz) as monomer into the corresponding conjugated microporous polymer PTPADTz by both $FeCl_3$-catalyzed chemical polymerization and electrochemical polymerization as cathode materials for Li–organic batteries. Distinct *in situ* color changes can be observed for the CMP film cathode during the charge/discharge process, due to doping and dedoping of the layers as this is accompanied by changing the oxidation state of the triphenylamine–bitriazine group. Later, Yu *et al.*[182] exploited the capability of a newly-synthesized redox-active donor-acceptor conjugated microporous polymer utilizing anthraquinone and benzene as linkers (AQ-CMP) as anodes for rechargeable air batteries. The working electrode was produced by loading the samples onto a piece of carbon cloth for macroscale shape formation. Such AQ-CMP materials exhibited remarkable redox capacity (202 mAh g^{-1} at 2 A g^{-1}) with good rate capability (58% capacity retention at 20 A g^{-1}) and cycling stability superior to their linear counterpart. Moreover, the various functionalities which can be introduce into CMPs allowed their usage for numerous areas, *e.g.* as protective layers in battery applications. Lai *et al.*[177] designed an electropolymerized CMP "nanoskin" for sulfur cathode protection in Li-S battery (**Figure 11c**). Upon sealing the sulfur cathode with the ultrathin CMP film, the active sulfur species and electrolyte are both encapsulated inside the electrode, which strengthened their retention abilities in batteries for high capacity (870 mAh g^{-1}) (**Figure 11d, e**) and long-term cycling (1000 cycles). Similarly, Loh *et al.*[176] prepared a solution-processable conjugated microporous thermo-setting polymer as the protective membrane layers for lithium anodes. The lithium anodes covered by such CMP materials showed a long-term stable performance during a test of several thousand hours. The superior stability (**Figure 11f**) was explained by the effective Li-ion diffusion ensured by nanofluidic nanochannels in the CMP layers.

4.2.3. Solar steam generation

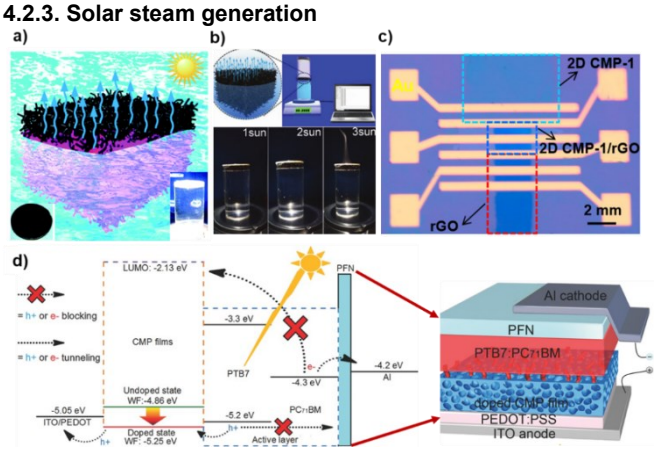


Figure 12. (a) Schematic of solar steam generation; (b) Schematic of solar steam generation and photographs of steam generated under varied solar irradiation energy of 1, 2, and 3 kW m^{-2} (1~3 sun); (c) Photograph of the 2D CMP-1/rGO heterostructure based organic vertical field-effect transistor; (d) Energy diagrams of PSCs using a CMP film as the anode interlayer. (a) Reproduced with permission. [95] Copyright 2018, Royal Society of Chemistry. (b) Reproduced with permission.[183] Copyright 2018,

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CMPs with different shapes have been engineered for photothermal conversion devices which gain importance for applications such as power generation, desalination or solar steam generation. Mu *et al.*[95] reported the first example of CMP aerogel-based photothermal materials, denoted as CMPA. This material was mechanically and chemically robust, elastic and processable, with a porosity as high as 94%. Remarkable solar steam generation efficiencies were achieved (88%) under different light intensities (**Figure 12a**). Later, the same group[183] further pyrolyzed these materials to afford carbon aerogels. The obtained carbon aerogels were superhydrophobic, highly porous, low thermal conductivity and exhibited wide light absorption. This material demonstrated a high solar steam generation efficiency of 86.8% under 1 sun irradiation (**Figure 12b**). On the basis of the previous research, Zhu *et al.*[147] further prepared nitrogen-rich CMPAs coated with a thin layer of polypyrrole to improve the light absorption. For solar steam generation, these aerogels achieved a high energy conversion efficiency of 80% under 1 kW m^{-2} illumination. Such reports thus provided effective pathways for the design and fabrication of macroscale CMP-based solar steam generators.

4.2.4. Photovoltaics

Ma *et al.*[185] reported a novel CMP film for the anode intermediate layer in a polymer solar cell (PSC). This electropolymerized TPTCz-CMP film exhibited a high specific surface area, remarkable work function and electrical conductivity, along with high lowest unoccupied molecular orbital (LUMO) energy level (-2.13 eV) which endows enough electron-blocking capability to act as anode interlayer/hole transporting layer in a PSC, benefiting its photovoltaic performance (**Figure 12d).** Hence, polymer solar cells built upon these CMP films achieved a high solar conversion efficiency of 8.42% *via* integrated optimization of the CMP film anode and cathode interlayers. Before, the record of power conversion efficiency (PCE) of the bulk-heterojunction (BHJ) PSCs has reached 8% (conventional device structure). Later, Jiang *et al.*[186] reported an electropolymerized polyborane carbazole-based thin film as a novel electrode interlayer. It was found that this conductive thin film exhibited unprecedented low work-function-selective electron flow. Upon ionization and electro-oxidation, the polyborane carbazole film was turned into a hole conductor with increased work function. In addition, these ionic films were compatible with different types of electrodes and could thus be assembled into various devices including solar cells and light-emitting diode devices, with high power conversion efficiency[101, 103, 187]. On the other hand, Huang and co-workers[184] proposed a universal surface-assisted oxidative polymerization strategy for preparing large-area ultrathin 2D CMPs on arbitrary substrates. The as-synthesized 2D CMPs (**Figure 12c**) with average thickness of 1.5-3.6 nm were applied in an organic vertical field-effect transistor, and exhibited typical p-type behavior with high reproducibility and on/off current ratio of 4.82 × 10^3.

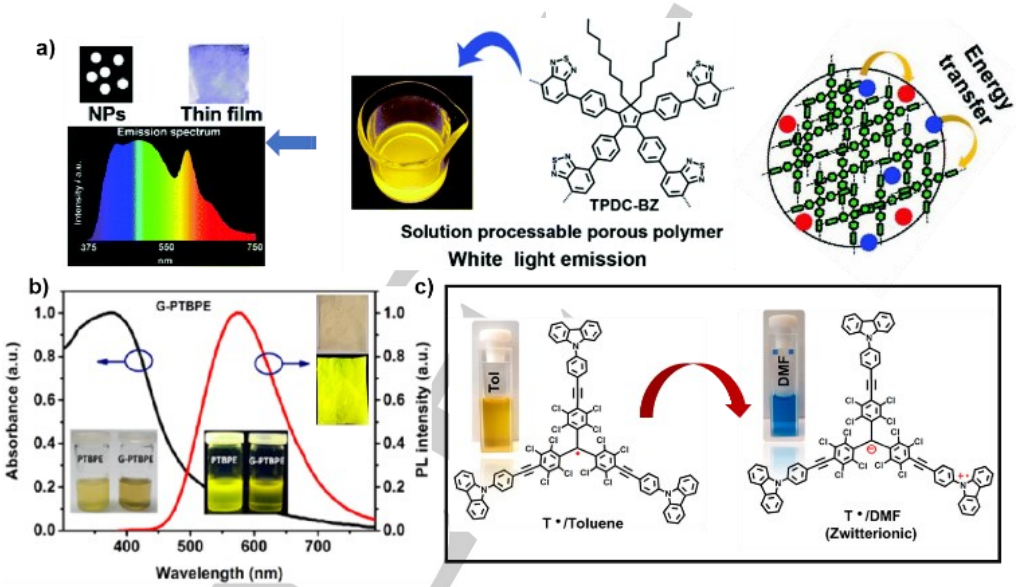

Figure 13. (a) Structure of strongly fluorescent and soluble porous organic polymer (TPDC-BZ) and its white light emission in solution, nanoparticles (NPs) and as thin film; (b) Normalized electronic absorption and fluorescence spectra of G-PTBPE; (c) Chemical formula representations of radicalic and zwitterionic resonance structures of T• in toluene and DMF (inset shows daylight images of T• in toluene and DMF). (a) Reproduced with permission.[188] Copyright 2017, Royal Society of Chemistry. (b) Reproduced with permission.[166] Copyright 2015, American Chemical Society. (c) Reproduced with permission.[189] Copyright 2019, American Chemical Society.

4.3. Light-emitting and sensing devices

CMP thin films often show fast and amplified signal response, and are therefore potentially useful for developing portable sensing devices. Aggregation-induced emission (AIE)[134] refers to the phenomenon in which molecular aggregates exhibit stronger emission than the single molecules. This can be utilized for a wide range of applications in the fields of light-emitting diodes,

fluorescent labels, light capture, photonic devices or chemical and biological sensing. Indeed several examples of porous polymers with AIE building blocks in their backbone have been reported and were fabricated into light-emitting and sensing devices. For instance, Jiang *et al*.[190] used tetraphenylethylene as AIE units and equipped this molecule with electropolymerizable carbazole groups (denoted as TPECz). After electropolymerization, the PolyTPECz-based thin CMP films exhibited fluorescence quenching response to 2,4,6-trinitriphenol (TNP) solutions (50 ppm), applied as model compound for explosives, with good selectivity, low detection limits and proper fluorescence retention after washing out the analyte. Afterwards, Patra *et al*.[188] *et al.* synthesized a strongly fluorescent CMP from tetrakis-(4-bromophenyl)-5,5-dioctylcyclopentadiene (TBDC) and 2,1,3-benzothiadiazole-4,7-bis(boronic acid pinacol ester) (BZ-BAPE). Most notable, due to attached long hydrocarbon chains and, probably, a lower polymerization degree, i.e. crosslinking density, this CMP (named TPDC-Bz) was still soluble in chloroform and thus solution processable. The as-synthesized CMP was thereafter fabricated into four different forms all showing white light emission: solutions, aqueous dispersion of nanoparticles, gels and transparent thin films (**Figure 13a**). Chen et al.[152] fabricated a light-emitting nanofiber CMP which used tetrakis(4-bromophenyl)ethene (TBPE) as monomer for the Yamamoto-type homocoupling into TPE-based polymer networks (PTBPE) by electrospinning CMP/poly(lactic acid) (PLA) mixtures. Moreover, using 4-iodophenyl-substituted graphene (RGO-IBz) as templates prepared graphene-PTBPE(G-PTBPE) in Yamamoto-type or Sonogashira−Hagihara coupling reactions. Both PTBPE and G-PTBPE emitted yellow light, compared to PTBPE, G-PTBPE of absorption band dispersion peaks at 380 nm, slightly blue-shifted and much weaker than PTBPE (**Figure 13b**). Upon excitation at 380 nm. Recently, Scherf *et al.* [189] further confirmed the application of CMP films in magnetoelectronic and sensor devices. A simple one-step electrochemical approach was applied to fabricate π-radicalic CMP thin films based on metal-free, electrochemical oxidative C−C coupling using carbazole substituted polychlorotriphenyl methane (PTM)-based radical monomers. After immersing the film into a n-Bu_4NOH solution, it was observed that the radical CMP film was reduced to the corresponding anionic CMPs, accompanied by an obvious color change noticable with the naked eyes. Notably, different electron paramagnetic resonance (EPR) curves and color changes can also be observed in toluene and DMF solutions (**Figure 12c**).

4.4. Filtration and purification

Notably, conjugated structures in CMPs exhibited strong affinity for aromatic/cationic pollutants[79, 80], and their rigid 3D carbon-carbon/carbon-nitrogen architectures showed superior chemical resistance towards organic solvents, which in turn made macroscale CMPs interesting for the adsorption/separation of organic contaminants.

4.4.1. Nanofiltration

Tang *et al.*[191] prepared CMP membranes with backbones composed of 1,3,5-triethynylbenzene (1,3,5-TEB) with three different dibromobenzenes, 1,4-dibromobenzene (1,4-DBB), 1,3-dibromobenzene (1,3-DBB) and 1,2-dibromobenzene (1,2-DBB) through Sonogashira–Hagihara polymerization. The CMP membranes with a thickness of 42 nm showed high resistance to organic solvents and excellent solute retention and solvent nanofiltration performance in both non-polar hexane and polar methanol, with permeabilities reaching 32 and 22 L m^{-2} h^{-1} bar^{-1}, respectively. Furthermore, the application of CMPs as organic solvent nanofiltration membranes were simulated by molecular simulations to construct realistic structural models to understand the origin of the high performance (**Figure 14a-f**) and to precisely adjust the aperture for ultrafast OSN. The pore-size distribution for each CMP polymer network could be derived from the above simulation results. The simulated increased pore size was consistent with a decreased rejection performance of *p*-CMP, *m*-CMP and *o*-CMP. In order to prepare robust CMP membranes, Lai *et al.*[192] applied electropolymerization for membrane synthesis inside a robust carbon nanotube scaffold. The as-prepared CMP membranes composed of a rigid carbazole monomer (SpCz, 2,2',7,7'-tetra(carbazol-9-yl)-9,9'-spirobifluorene) showed superior mechanical strength and ductility, and were utilized for high-performing organic solvent nanofiltration. A stable methanol permeance of 28 L m^{-2} h^{-1} bar^{-1} was achieved for the CMP membrane, which is superior to most reported organic solvent nanofiltration membranes.[193, 194]

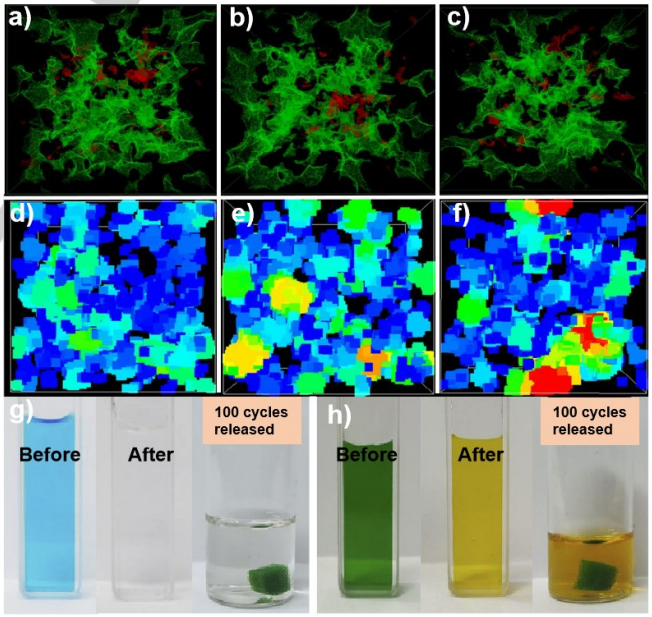

Figure 14. Interconnected (green) and disconnected (red) voids in (a) p-CMP, (b) m-CMP, (c) o-CMP and polyamide with respect to a probe of 1.2 Å radius, which represents the largest probe size that can diffuse across any of the rigid models; Voids colored with respect to pore diameter for (d) p-CMP, (e) m-CMP and (f) o-CMP; (g) Photographs of a methylene blue aqueous solution (5.0×10^{-5} M) before and after adsorption; (h) Photographs of the aqueous solution before and after adsorption. (a-f) Reproduced with permission.[191] Copyright 2018, Springer Nature. (g, h) Reproduced with permission.[195] Copyright 2017, Elsevier.

4.4.2. Purification

Chang *et al.*[195] prepared a compressible multilayer CMP composite *via* Sonogashira-Hagihara coupling reaction of 1,3,5-triethynyl benzene, 1.4-diiodobenzene and 2,5-diiodobenzoic acid inside a polyurethane sponge. The CMP formed inside the polyurethane sponge network demonstrated a fiber-shaped morphology and can be deprotonated to a carboxylate polymer with KOH. The resulting polymer composite can be manually compressed and released in an aqueous solution of methylene blue to quickly remove the cationic dyes and recycle it afterwards (**Figure 14g, h**). Li *et al.*[196] reported the synthesis of a superhydrophobic conjugated microporous polymer composed of 1,3,5-triethynylbenzene (TEB) and 2-amino-3,5-dibromo pyridine with two aromatic heterocyclic isomers bearing the $-NH_2$ functional group. This aminopyridine-containing CMP (APCMP-1, 2, 3, 4) exhibited good selectivity, fast adsorption kinetics, excellent recyclability and absorbency for a wide range of organic solvents and oils. The adsorption capacity of APCMP-1 for four type of oils (pumping oil, kerosene, diesel oil and gasoline) varied at the range of 1380–3280 wt%. After the CMP was coated onto a sponge material, the formed CMP sponge composite was applied for the absorption of various organic solvents and oils. The same group[197] further attempted to turn APCMP into an aerogel by sol-gel transformation. The hierarchically porous structure of the APCMP aerogel benefited the filtration especially for air-floating particles with different sizes. The capture efficiency of PM2.5 and PM10 (PM 2.5/10 means the mass per cubic meter of air of particles with a size (diameter) generally less than 2.5/10 μm) were respectively up to PM2.5 ≥ 99.57 ± 0.19% and PM10 ≥ 99.98 ± 0.01% in a long-term durability test. These values are higher than can be achieved for fibrous high-efficiency filters.[198-200] Additionally, the APCMP aerogels could also easily be regenerated. In further studies, two tubular CMPs (termed as CMP-As and CMP-Bs) synthesized from 1,3,5-triethynylbenzene and brominated monomers including 1,4-dibromobenzene and 4,4-dibromobiphenyl, respectively, were studied.[201] Both CMPs presented excellent performance of particulate matter filtration and oil/water separation. The filtration efficiency of such CMPs exceeded 99.15% for PM 2.5 and PM10 at room temperature and high humidity (RH > 93 %). These studies show, that CMPs are indeed promising materials for further explorations, *e.g.* in oil-oil separation, microplastics adsorption, organic solvent dialysis, thus a wide variety of the filtration and purification applications. It can be surely argued that, for real-life filtration/purification systems large amounts of the absorber material are needed, and that therefore the relatively high price of CMPs compared, for example to activated carbons or other typical filter/absorber materials, make such an application unrealistic for CMPs. Especially for this reason, macroscale CMPs prepared by thin coatings of CMPs on another, much cheaper material, as exemplified in the works above, could be a solution, as the separation performance provided by the CMP is achieved with lowest material amounts.

4.5. Heterogeneous catalysis

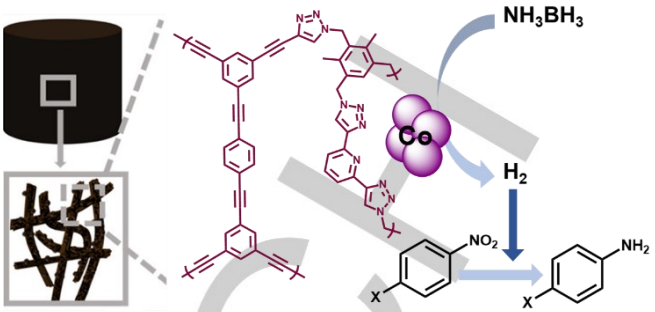

Figure 15. Schematic illustration of the tandem dehydrogenation of ammonia borane and hydrogenation of nitro compounds. Reproduced with permission.[202] Copyright 2018, American Chemical Society. Further permissions related to the material excerpted should be directed to the ACS (https://pubs.acs.org/doi/10.1021/acsomega.8b01416).

Due to the versatile functional groups which can be introduced into CMPs, they can be easily equipped with catalytic active sites, which are then fully accessible through the rich micro-/mesoporosity. With macroscopic architectures, CMP-based catalyst are more easily processable and recyclable and can especially be applied in flow reactors, where very fine powders would cause several problems[203, 204]. For example, Lee *et al.* prepared a CMP membrane embedded with silver nanoparticles through the *in-situ* Sonogashira-Hagihara coupling reaction of 1,3,5-triethynylbenzene and 1,4-diiodobenzene using an electrospun nanofiber membrane as a sacrificial template[205]. This catalytic CMP membrane was used for the catalytic reduction reaction of 4-nitrophenol and recycled 3 times without loss in catalytic activity. Alternatively, by performing the Sonogashira-Hagihara reaction between 1,4-diiodobenzene and 1,3,5-triethynylbenzene in a gel state, Chang *et al.*[202] successfully prepared a series of compressible CMP monoliths which were further functionalized by the click reaction with 1,3,5-tris(azidomethyl)-2,4,6-trimethylbenzene and 2,6-diethynylpyridine. Cobalt nanoparticles were generated *in-situ* on the CMP monoliths, which were then applied in a tandem reaction involving the dehydrogenation of ammonia borane and subsequent hydrogenation of nitro compounds (**Figure 15**). Notably, such CMP monoliths exhibited morphologies of fiber-shaped nano-bundles intercalated with micron-sized spherical polymer particles, where catalytic active cobalt nanoparticles can be dispersed evenly. Therefore, the speed of the tandem reaction can be greatly accelerated when the catalyst was compressed and released during the reaction, because the repeated compression and release facilitated interfacial contact between the reactants and active sites (cobalt nanoparticles) in CMP monoliths. Furthermore, the same group[206] reported a compressible polyimide composite with Pd binding sites prepared by *in-situ* polymerization in a melamine sponge. After coordinated with Pd(II) ions, this hierarchical porous CMP sponge was applied for heterogeneous Suzuki coupling reactions. Consequently, such Pd incorporated CMP sponges were treated as a promising material platform as for heterogeneous catalysis.

4.6. Other applications

Thermal insulation. Li *et al*.[207] synthesized a monolithic CMP composed of entangled nanotubes (M-CMPsN) *via* Sonogashira-Hagihara coupling reaction. The M-CMPsN exhibit promising mechanical properties with a compressive strength of 92 kPa at 70% stain. Due to the porous structure, it exhibited thermal insulation with a thermal conductivity of only 0.03 Wm^{-1} K^{-1}. Compared with usually brittle organic thermal insulation materials and powdery CMPs, the elastomeric M-CMPsN was flexible and mechanical robust and in turn could be engineered into different shapes for practical applications.

Antibacterial materials. Lei *et al*.[73] synthesized a novel conjugated microporous polymer (I-CMPs) from trichloroisocyanuric acid and ethynylbenzene as building blocks, *via* Sonogashira-Hagihara cross-coupling condensation reaction. The specific surface areas of I-CMP-1 and I-CMP-2 were measured to be 383 m^2 g^{-1} and 246 m^2 g^{-1}, respectively. The as-synthesized I-CMP-1 and I-CMP-2 monolithic nano-sponges fabricated *via* the sol-gel method exhibited superhydrophobicity with a water contact angle of 151° and 158°. The combined effects of covalently incorporated isocyanurate, abundant pores, high specific surface area together with I-CMPs can provide many accessible active sites for contacting and killing bacteria. The minimum inhibitory concentration values of I-CMP-1 and I-CMP-2 are 50 and 100 $\mu g \cdot mL^{-1}$ for E. coli, and 100 and 250 $\mu g \cdot mL^{-1}$ for S. aureus per hour, respectively. In addition, I-CMPs were processed into sponges or free-standing films, which further broadens the application of I-CMPs in antibacterial aspects.

Resistive memory. Huang *et al*.[208] designed and constructed a resistive memristor based on a large area 2D CMP film synthesized from surface-confined polymerization, featuring rewritable nonvolatile memory with a low set voltage in a graphene/CMP/metal diode. By assembling the solution-processable all-polymer one diode–one resistor (1D1R) cross-bar memory array by directly stacking the CMP memristor with a poly(3-hexylthiophene) rectifier, the sandwich structured CMP device exhibited nonvolatile flash performance, with features of low set voltage (~1.65 V), stable cycling endurance (>25 cycles), and long retention time (10^4 s). Subsequently, a bilayer heterojunction was fabricated by a two-terminal diode with CMP layer as resistive switching medium[209]. This bilayer heterojunction device posed flash feature with a high on/off ratio (>10^5) and a long retention time (>3.0×10^4 s), which exceeds most of reported conjugated polymers memories. These works presented an effective strategy for the precise and facile preparation of bilayer heterostructure conjugated polymer films for integrated polymer memory electronics.

6. Conclusion and outlook

Macroscale conjugated microporous polymers represent a future multifunctional platform for broadening the applicability of CMPs in many areas. *Via* combining π-conjugation (providing light absorption and semiconductivity), meso-/microporosities (large surface area and fast mass transport) with macroscopic architectures (flexibility and processability within devices and reactors), CMPs can be vastly expanded into applications such as organic photovoltaics[103], nanofiltration[210], solar steam generation[211], gas separation[212] and energy storage devices[213]. Compared to conventional synthetic techniques applied for CMPs, specific methods such as electropolymerization[103, 214], template synthesis[215], post-synthetic modification[216], mechanochemical processes[170], and others are emerging for creating CMPs with control of length scales from nanometers to centimeter/meter macrostructures. Such macroscale structures can provide access to diverse forms and patterns (membranes, thin films, aerogels, sponges, *etc.*). Nevertheless, structure control on a macroscale is still challenging for CMPs. For example, macroscale fiber-shaped CMPs are barely reported until the first example very recently[81], and more sophisticated structures of macroscale CMPs such as 3D-printed ones would be of immense interest. There is indeed much room for further developing new synthetic concepts to modulate pore size and surface functionality at multiple length scales and generate new functions in particular.

Macroscale CMPs in versatile shapes and forms, might enable purification and gas separation techniques. Indeed, the filtration/separation area has been a rising field for macroscale CMPs, where the nanoconfinement plays a pivotal role in controlling the permeation and adsorption. Energy applications based on macroscale CMPs are also rapidly expanding, such as solar steam generation or battery membranes. In terms of catalysis, CMPs in macroscale can create an adaptive pore environment to individual reaction by tuning the nanoconfinement effect, as well as maintaining the macroscopic architectures as the substrate and reactor.

As an interdisciplinary topic the generation of macroscale CMPs combine diverse synthetic methodologies. For example, the combination of CMPs with polymer processing technologies and porous polymer chemistries, *e.g.*, 3-D printing[217], microfluidics[218], wet/electro-spinning[219], composite blending[220] and molding[221], to name just a few, brings exciting opportunities. With more efforts along this line, we can foresee that macroscale CMPs can generate novel products for industrial applications.

Acknowledgements

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Conflict of interest

The authors declare no conflict of interest.

Keywords: macroscale conjugated microporous polymers • adsorption and separation • energy storage and conversion • heterogeneous catalysis • smart materials

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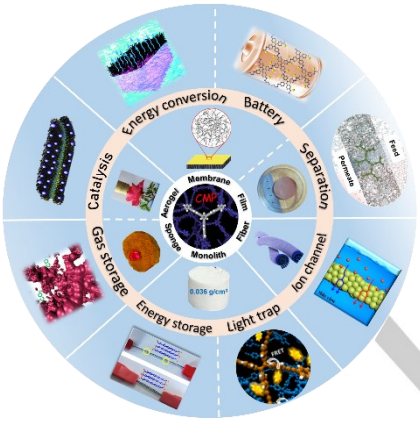
REVIEW

Recent progress in macroscale conjugated microporous polymers for gas adsorption, energy, sensing, filtration/purification and catalysis applications are reviewed, with particular focus on how the macroscopic architectures brings about more functionalities and opportunities.



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Macroscale conjugated microporous polymers: controlling versatile functionalities over several dimensions

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